JGi JAIN DEEMED-TO-BE UNIVERSITY | FACULTY OF ENGINEERING AND TECHNOLOGY

Department of Information Science and Engineering

PCL Review PROGRESS REPORT		
Batch No	**16**	
Guide	**Dr. Chayapathi A R**	
PCL Project Title	**Range optimization of sensors**	
Review No	**1st review**	
Date of Submission	**18th November, 2025**	
Date	**From:6th October , 2025**	**To: 17th November , 2025**

Sl. No.	Student Name	USN	Signature with date
1	Smriti Singh	**23BTRIS040**	
2	Jeffrey Thomas	**23BTRIS016**	

Remarks by Guide

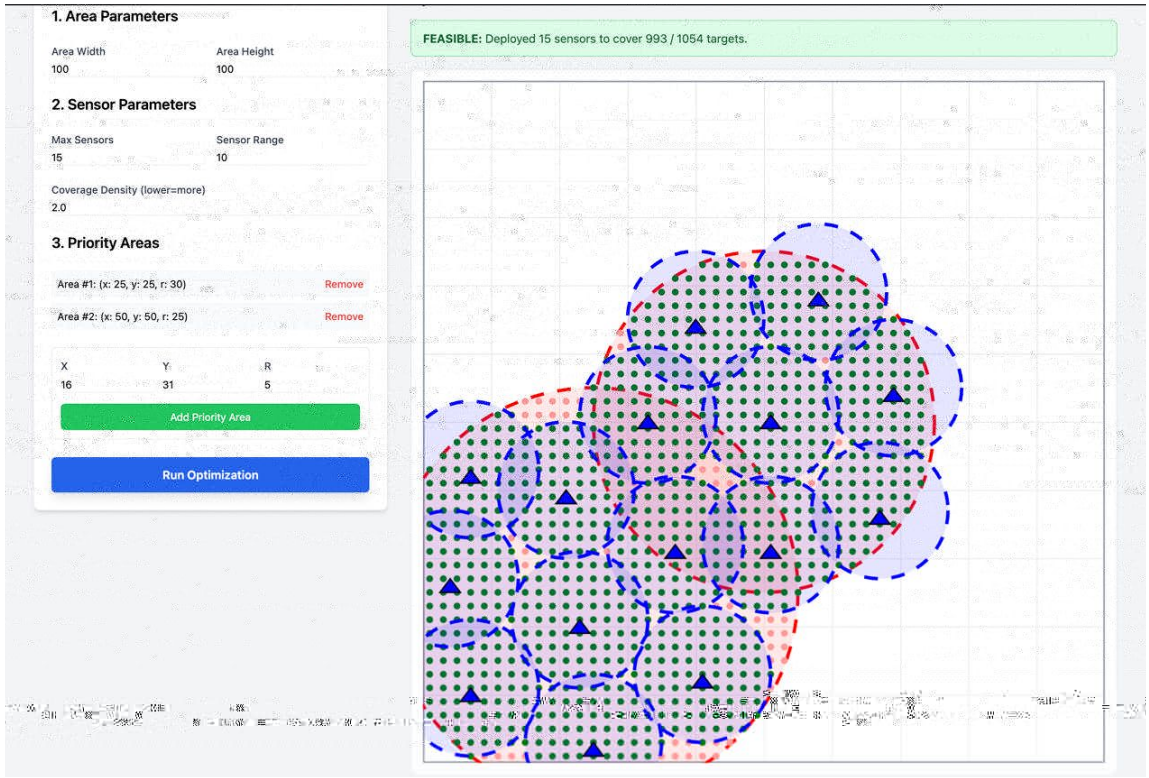
Progress: (Describe actual progress of current Review).

The project has finally achieved its objectives. The current version successfully implements an area-based sensor optimization model using Python and Google OR-Tools (CP-SAT solver) and displays them on a fully functional web-app.

Key milestones achieved include:

A fully functional web app ready for demonstration, showcasing our project in an interactive manner.

- Users can input area size, priority zones, and sensor parameters.
- Deployed sensors, coverage circles, and covered/uncovered sub-targets will be displayed clearly on an interactive canvas.
- Users can experiment with sensor ranges and see the effect on coverage instantly.
- Backend optimization logic will handle different scenarios efficiently and return results quickly.



Demo of the working model in development mode

Work schedule and expected results for the next Review

The upcoming work focuses on deploying the web application, validating system performance, and completing the thesis documentation. Although the application is fully functional, the deployment stage remains pending.

1. **Web App Deployment**

The backend and frontend will be configured for cloud deployment using a suitable hosting platform. Compatibility testing across devices and browsers will be carried out. The expected outcome is a stable, publicly accessible web application.

2. UI/UX Refinements

Minor improvements will be made to the visualization interface to enhance clarity, including better labeling and responsiveness.

3. Thesis Paper Preparation

The thesis will be drafted, covering methodology, system design, algorithmic approach, results, and analysis. Figures, flowcharts, and references will be added.

References:

1. Smith, J., & Kumar, R. (2022). Optimal Sensor Deployment Strategies in Wireless Networks. International Journal of IoT Research, 15(4), 45–58.
2. Li, Y., & Chen, H. (2021). *Constraint Programming Approaches for Coverage Optimization*. IEEE Transactions on Computational Intelligence, 12(3), 102–115.
3. Google OR-Tools Documentation. (2023). Retrieved from https://developers.google.com/optimization
4. Gupta, P., & Verma, S. (2020). Wireless Sensor Networks: Design and Applications. Springer, New York.

Guide	PCL Coordinator	HOD-ISE
(Signature with Date)	(Signature with Date)	(Signature with Date)